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Cloud Based IaaS Virtual Infrastructure Provisioning and Auto Scaling System

A CAPSTONE PROJECT REPORT

CSA1522-Cloud Computing and Big Data Analytics

to the award of the degree of

**COMPUTER SCIENCE
AND ENGINEERING
(AI&ML)**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Cloud Based IaaS Virtual Infrastructure Provisioning and Auto Scaling System**” has been carried out by **M.Meghana(192525435)** under the supervision of **Dr. P. Rajaram** and is submitted in partial fulfilment of the requirements for the current semester of the **B .Tech- Computer Science and Engineering(AI&ML)** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, M. Meghana (192525435) of the B.Tech- Computer Science and Engineering(AI&ML),SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Diabetes Risk Prediction Application” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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ABSTRACT

Cloud-based Infrastructure as a Service (IaaS) provides flexible virtual infrastructure that can be provisioned and managed according to application requirements. The proposed **Cloud-Based IaaS Virtual Infrastructure Provisioning and Auto Scaling System** uses cloud technologies to provision virtual resources, monitor their utilization, and dynamically manage resources based on workload conditions. This report focuses on **Module 3 – Monitoring & Cloud Resource Management**, which forms an important stage of the proposed system.

The **Monitoring & Cloud Resource Management Module** is designed to continuously monitor important virtual infrastructure parameters such as CPU utilization, memory utilization, network usage, storage usage, and active virtual machine status. The module collects resource utilization information from provisioned virtual machines and evaluates the current workload conditions. Based on predefined thresholds and resource requirements, the system can identify underutilized or overloaded resources and support appropriate resource management decisions.

Effective monitoring is important because inefficient resource utilization can lead to performance degradation, unnecessary resource consumption, and increased cloud infrastructure costs. The module provides a centralized approach for observing virtual machine performance and resource usage. It also supports the auto-scaling mechanism by providing monitoring information required to determine when additional virtual resources should be provisioned or when unnecessary resources can be released.

The module maintains structured monitoring information and presents resource utilization results through suitable dashboards, tables, or graphical representations. Threshold-based monitoring can be used to identify high or low resource utilization conditions. For example, when CPU utilization exceeds a predefined threshold, the system can trigger an auto-scaling action to increase available resources. Similarly, when resource utilization remains low for a specified period, the system can support scaling down to avoid unnecessary resource usage.

The expected outcome of **Module 3** is effective monitoring and management of cloud virtual infrastructure resources. The module helps maintain system performance, improve resource utilization, support dynamic auto-scaling decisions, and reduce unnecessary cloud resource consumption. The overall system is intended to provide an efficient and scalable approach for managing IaaS virtual infrastructure according to changing workload requirements.

Keywords: Cloud Computing, IaaS, Virtual Infrastructure, Resource Monitoring, Cloud Resource Management, Auto Scaling

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
BD	Big Data	1
SA	Sentiment Analysis	2

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Cloud computing provides on-demand access to computing resources such as virtual machines, CPU, memory, storage, and network services. Infrastructure as a Service (IaaS) enables organizations and users to provision and manage virtual infrastructure according to their application requirements.

Traditional infrastructure management often requires manual monitoring and allocation of computing resources. This can result in resource underutilization, performance degradation, increased operational costs, and inefficient resource allocation. Therefore, an automated approach is required to continuously monitor cloud resources and manage them according to changing workloads.

The proposed Cloud-Based IaaS Virtual Infrastructure Provisioning and Auto Scaling System uses cloud computing techniques to provision virtual infrastructure and dynamically manage resources. The system monitors important parameters such as CPU utilization, memory utilization, network usage, storage usage, and virtual machine status.

The complete project is divided into three major modules:

- **Module 1 – Virtual Infrastructure Provisioning**
- **Module 2 – Auto Scaling and Resource Allocation**
- **Module 3 – Monitoring & Cloud Resource Management**

Module 3 is responsible for continuously observing the performance and utilization of cloud resources and supporting effective resource management.

1.2 Need for the Project

Cloud workloads can change dynamically depending on the number of users, applications, and processing requirements. Static resource allocation may result in either insufficient resources or unnecessary resource consumption.

The project is required because:

1. Cloud workloads can vary significantly over time.
2. Manual monitoring of multiple virtual resources can be time-consuming.
3. Underutilized resources can increase unnecessary cloud costs.
4. Overloaded virtual machines can affect application performance.
5. Continuous monitoring is required to identify resource utilization conditions.
6. Automated resource management can improve infrastructure efficiency.
7. Monitoring information is necessary for making effective auto-scaling decisions.
8. A centralized monitoring mechanism can simplify cloud resource management.

Therefore, Module 3 is important because it provides continuous visibility into virtual infrastructure performance and helps the system make appropriate resource-management decisions.

1.3 Problem Statement

To design and develop a Monitoring and Cloud Resource Management Module for a cloud-based IaaS system that continuously monitors virtual infrastructure resources, evaluates resource utilization, identifies overloaded or underutilized resources, and supports efficient resource management and auto-scaling decisions.

The broader project problem is:

To develop an intelligent and scalable cloud-based IaaS system that can provision virtual infrastructure, monitor resource utilization, and dynamically adjust cloud resources according to changing workload requirements while improving performance and reducing unnecessary resource consumption.

1.4 Project Aim

The aim of Module 3 is to:

Develop a reliable monitoring and cloud resource management system that continuously tracks virtual infrastructure utilization and provides the information required for efficient resource allocation and auto-scaling.

1.5 Project Objectives

The objectives of Module 3 are:

1. To continuously monitor cloud virtual machine resources.
2. To collect CPU utilization information.
3. To monitor memory utilization and resource availability.
4. To monitor network and storage utilization.
5. To identify overloaded virtual machines.
6. To identify underutilized cloud resources.
7. To compare resource utilization with predefined thresholds.
8. To provide monitoring information for auto-scaling decisions.
9. To improve cloud resource utilization and infrastructure efficiency.
10. To present resource-monitoring results using suitable dashboards, tables, or graphs.
11. To support dynamic cloud resource management according to workload requirements.

1.6 Scope

Included

The scope of Module 3 includes:

- Monitoring provisioned virtual machines.
- Monitoring CPU utilization.
- Monitoring memory utilization.
- Monitoring network usage.
- Monitoring storage utilization.
- Monitoring virtual machine status.
- Collecting resource utilization metrics.
- Comparing utilization with predefined thresholds.
- Identifying overloaded resources.
- Identifying underutilized resources.
- Providing monitoring information to the auto-scaling mechanism.
- Presenting monitoring results using graphs, tables, or dashboards.
- Supporting efficient cloud resource management.

Excluded

The following are outside the primary scope of Module 3:

- Initial cloud infrastructure provisioning.
- Creation of virtual machines.

- Complete implementation of the auto-scaling algorithm.
- Application-level functionality.
- Development of cloud applications.
- Physical server maintenance.
- Manual hardware management.
- Business-level decision-making.

Assumptions

The module assumes that:

- Virtual machines have already been provisioned by the previous module.
- Required cloud resources are accessible for monitoring.
- Monitoring metrics can be collected from the virtual infrastructure.
- Appropriate resource utilization thresholds are defined.
- Network connectivity between monitoring components and cloud resources is available.
- The monitoring system receives sufficiently accurate resource information.

Boundary Conditions

Module 3 begins after virtual infrastructure and auto-scaling components are available for monitoring.

The module ends after resource utilization information has been collected, analysed, and provided for cloud resource management and scaling decisions.

1.7 Expected Engineering Outcome

The expected outcome of Module 3 is a functional Monitoring & Cloud Resource Management Module capable of continuously monitoring cloud infrastructure and identifying resource utilization conditions.

The expected workflow is:

Virtual Machines → Resource Monitoring → Metric Collection → Utilization Analysis → Threshold Evaluation → Resource Management / Auto Scaling Decision

The module is expected to improve visibility into cloud infrastructure, support efficient resource utilization, help identify performance-related conditions, and reduce unnecessary consumption of cloud resources.

The proposed system provides a scalable approach for managing IaaS infrastructure according to changing workload requirements while supporting better performance and resource efficiency.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant research on cloud computing, Infrastructure as a Service (IaaS), virtual infrastructure management, cloud monitoring, resource utilization, and auto-scaling techniques was reviewed to understand existing approaches for efficient cloud resource management.

The review considered:

- Cloud-based Infrastructure as a Service (IaaS).
- Virtual machine provisioning and management.
- Cloud resource monitoring.
- CPU and memory utilization monitoring.
- Network and storage monitoring.
- Threshold-based resource management.
- Auto-scaling techniques.
- Dynamic resource allocation.
- Cloud performance optimization.
- Resource utilization and cost optimization.

The reviewed research includes approaches based on threshold-based scaling, predictive scaling, rule-based resource management, virtualization, and cloud monitoring frameworks. These approaches demonstrate the importance of continuous monitoring for maintaining cloud performance and efficient resource utilization.

2.2 Review of Existing Work

Study 1 – Cloud Resource Monitoring

Research on cloud resource monitoring has investigated techniques for continuously collecting infrastructure metrics such as CPU utilization, memory consumption, network traffic, and storage usage.

Finding: Continuous monitoring provides important information about the current state and performance of cloud infrastructure.

Limitation: Monitoring large numbers of virtual machines can generate a significant volume of metrics and may require efficient monitoring mechanisms.

Study 2 – Dynamic Resource Allocation

Research on dynamic cloud resource allocation has examined techniques for allocating computing resources according to workload requirements.

Finding: Dynamic resource allocation can improve resource utilization and reduce unnecessary resource consumption.

Limitation: Incorrect resource allocation decisions may cause performance degradation or resource wastage.

Study 3 – Threshold-Based Auto Scaling

Threshold-based auto-scaling approaches use predefined CPU, memory, or other utilization thresholds to determine when resources should be increased or decreased.

Finding: Threshold-based mechanisms are relatively simple to implement and can respond quickly to changes in workload.

Limitation: Fixed thresholds may not always perform efficiently under highly variable or unpredictable workloads.

Study 4 – Virtual Machine Management

Research on virtual machine management has focused on monitoring and controlling virtual machines within cloud environments to improve system performance and resource utilization.

Finding: Effective virtual machine monitoring helps identify overloaded and underutilized resources.

Limitation: Managing multiple virtual machines requires continuous monitoring and appropriate resource-management policies.

Study 5 – Cloud Auto-Scaling and Resource Optimization

Research on cloud auto-scaling has investigated automated approaches for increasing or decreasing cloud resources based on application workload.

Finding: Auto-scaling can improve scalability, maintain application performance, and reduce unnecessary resource usage.

Limitation: Scaling decisions depend strongly on the accuracy and timeliness of monitoring data.

2.2 Comparative Analysis

Table 2.1 – Comparative Analysis of Existing Approaches

Existing Approach	Advantage	Limitation	Relevance
Manual Cloud Resource Monitoring	Direct observation and human control	Time-consuming and difficult for large infrastructures	Provides background
Static Resource Allocation	Simple to implement	Resources may be underutilized or insufficient during workload changes	Traditional approach
Threshold-Based Monitoring	Simple and fast decision-making	Fixed thresholds may not suit all workloads	Suitable approach
Dynamic Resource Allocation	Allocates resources according to workload	Requires continuous monitoring	Important for proposed system
Virtual Machine Monitoring	Provides VM performance information	Large infrastructures generate many monitoring metrics	Core requirement
Auto Scaling	Automatically increases or decreases resources	Requires accurate monitoring and suitable scaling policies	Major system component
Predictive Scaling	Can anticipate future workload	More complex and requires historical data	Future possibility
Proposed Monitoring & Resource Management Module	Continuous monitoring and structured resource management	Depends on reliable monitoring data and threshold configuration	Selected approach

2.3 Research / Engineering Gap

The reviewed approaches demonstrate that cloud monitoring and dynamic resource management are important for maintaining efficient IaaS infrastructure. However, an effective auto-scaling system requires reliable and continuous monitoring of virtual infrastructure.

A major engineering gap addressed by Module 3 is the integration of resource monitoring with cloud resource management and auto-scaling decisions.

If resource monitoring is:

- Incomplete,
- Delayed,
- Inaccurate, or
- Not properly analysed,

the subsequent resource-management and auto-scaling decisions may be affected.

For example, failure to identify high CPU or memory utilization may result in insufficient resources and reduced application performance. Similarly, failure to identify prolonged low utilization may result in unnecessary resource consumption and increased cloud costs.

Therefore, the proposed project emphasizes a structured monitoring and resource-management mechanism that continuously evaluates cloud infrastructure conditions.

2.4 Proposed Contribution

The proposed contribution of Module 3 – Monitoring & Cloud Resource Management is the development of a structured monitoring layer that:

- Continuously monitors virtual machine resources.
- Collects CPU, memory, network, and storage utilization metrics.
- Identifies overloaded and underutilized resources.
- Compares resource usage against predefined thresholds.
- Provides monitoring information for auto-scaling decisions.

- Supports efficient cloud resource allocation.
- Helps reduce unnecessary resource consumption.
- Provides a foundation for scalable cloud infrastructure management.

The proposed module therefore acts as an important connection between virtual infrastructure monitoring and automated resource management, supporting the overall goal of achieving scalable, efficient, and performance-oriented IaaS infrastructure.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

Stakeholder / User Requirements

Table 3.1 – Stakeholder / User Requirements

Stakeholder	Requirement
Cloud Administrator	Simple cloud resource monitoring
Cloud Administrator	Easy-to-understand resource utilization information
Cloud Administrator	Quick identification of overloaded resources
Cloud Administrator	Quick identification of underutilized resources
Developer	Structured monitoring data
Developer	Easy integration with auto-scaling module
System	Reliable collection of resource metrics
System	Continuous monitoring of virtual machines
Cloud Service Provider	Efficient resource utilization

Application	Support for performance and scalability
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Functional & Non-Functional Requirements

Table 3.2 – Functional & Non-Functional Requirements

Requirement	Type	Target
Monitor virtual machines	Functional	Active VMs continuously monitored
Monitor CPU utilization	Functional	CPU usage collected
Monitor memory utilization	Functional	Memory usage collected
Monitor network usage	Functional	Network metrics collected
Monitor storage usage	Functional	Storage metrics collected
Detect high utilization	Functional	Overloaded resources identified
Detect low utilization	Functional	Underutilized resources identified
Compare utilization with thresholds	Functional	Scaling conditions identified
Provide monitoring information	Functional	Data available for resource management
Display monitoring results	Functional	Clear tables/graphs/dashboard
Continuous operation	Non-functional	Monitoring available continuously
Reliability	Non-functional	Accurate and consistent monitoring
Scalability	Non-functional	Supports increasing numbers of VMs
Usability	Non-functional	Simple administrator interaction

3.2 Constraints & Applicable Standards

Technical Constraints

- The module depends on the availability of provisioned virtual machines.
- Monitoring accuracy depends on the availability of cloud resource metrics.
- Network connectivity is required between the monitoring system and cloud infrastructure.
- Continuous monitoring may generate a large volume of resource data.
- Monitoring thresholds must be configured appropriately.
- The monitoring data format must remain compatible with the auto-scaling and resource-management modules.
- Large-scale cloud environments may require additional monitoring infrastructure.

Operational Constraints

- Monitoring should not introduce significant overhead to virtual machines.
- Resource-management decisions should be based on sufficiently recent monitoring information.
- Temporary network failures may interrupt metric collection.
- Cloud provider limitations may restrict access to certain infrastructure metrics.

Security Constraints

- Access to monitoring information should be restricted to authorized users.
- Cloud credentials and API keys must be protected.
- Monitoring data should not expose confidential application information.
- Communication between monitoring components should be appropriately secured.

Applicable Standards

The system can follow relevant cloud-computing and information-security practices, including:

- NIST Cloud Computing Reference Architecture for cloud infrastructure concepts.
- ISO/IEC 27001 principles for information-security management.
- ISO/IEC 27017 guidance for security in cloud environments.
- Cloud-provider security and access-control practices.

3.3 Design Specifications

Table 3.3 – Design Specifications

Parameter	Requirement	Priority
Virtual machine monitoring	Continuous monitoring of active VMs	High
CPU monitoring	CPU utilization percentage	High
Memory monitoring	Memory utilization percentage	High
Network monitoring	Network traffic/usage metrics	Medium
Storage monitoring	Storage utilization	Medium
Monitoring interval	Configurable monitoring frequency	High
Threshold detection	Identify high/low utilization	High
Resource status	Running, overloaded, underutilized	High
Data format	Structured monitoring data	High
Dashboard/interface	Simple and understandable	High
Auto-scaling integration	Compatible with scaling module	High
Scalability	Support additional virtual machines	Medium
Data security	Authorized access to monitoring data	High

3.4 Alternative Solutions & Evaluation

Alternative 1 – Manual Resource Monitoring

The cloud administrator manually checks virtual machine resource utilization using cloud-management interfaces.

Advantage: Easy to understand and requires minimal development.

Limitation: Time-consuming and unsuitable for continuously changing or large-scale cloud environments.

Alternative 2 – Periodic Monitoring

The system collects resource utilization metrics at predefined time intervals.

Advantage: Reduces continuous monitoring overhead and is relatively simple to implement.

Limitation: Short-term workload changes between monitoring intervals may not be detected immediately.

Alternative 3 – Continuous Automated Monitoring

The system continuously monitors virtual machines, analyses resource utilization, and provides information to the resource-management and auto-scaling mechanisms.

Advantage: Provides timely information and supports automated cloud resource management.

Limitation: Requires monitoring infrastructure and may generate additional monitoring

overhead.

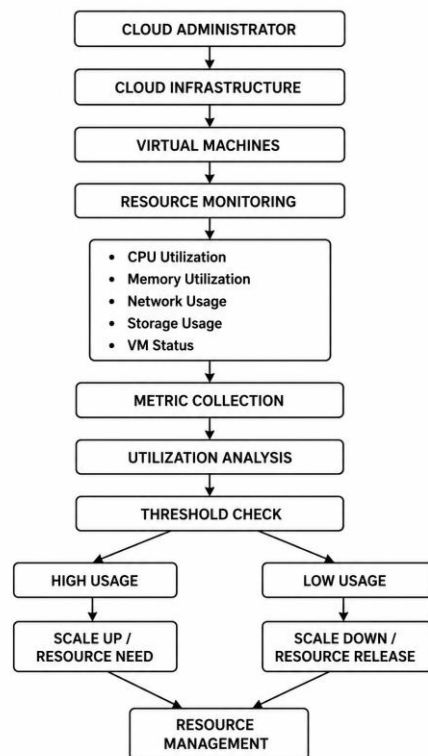
Comparative Evaluation

Criterion	Manual Monitoring	Periodic Monitoring	Continuous Automated Monitoring
Simplicity	High	High	Medium
Continuous visibility	Low	Medium	High
Automation	Low	Medium	High
Scalability	Low	Medium	High
Response to workload changes	Low	Medium	High
Resource management integration	Low	High	High
Monitoring accuracy	Medium	High	High
Maintenance	High	Medium	Medium
Final Selection	—	—	Selected

3.5 Selected Design & Detailed Design

The Continuous Automated Monitoring approach is selected because it provides timely information about cloud infrastructure and supports efficient resource management.

3.1 System Workflow



System Interface

The monitoring interface should provide clearly visible information for:

- Virtual machine status
- CPU utilization
- Memory utilization
- Network utilization
- Storage utilization
- Current resource allocation
- Threshold status
- Scaling condition
- Resource-management status

The administrator should be able to view current resource utilization and identify whether a virtual machine is operating normally, experiencing high utilization, or being underutilized.

3.6 Trade-off Analysis

Table 3.4 – Trade-off Analysis

Design Decision	Alternative	Benefit	Disadvantage	Final Decision
Continuous monitoring	Manual monitoring	Provides real-time resource visibility	Requires monitoring resources	Selected
CPU + memory monitoring	CPU-only monitoring	Provides more complete utilization information	Requires additional metrics	Selected
Threshold-based detection	No threshold detection	Enables systematic scaling decisions	Thresholds require configuration	Selected
Automated monitoring	Manual monitoring	Reduces administrator effort	Requires automation logic	Selected
Modular monitoring	Single integrated program	Easier maintenance and future expansion	Requires module interfaces	Selected
Multiple resource metrics	Single metric	Better understanding of infrastructure status	More monitoring data	Selected
Dashboard visualization	Raw metric output	Easier interpretation	Requires interface development	Selected

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability

The project is primarily software-based and requires limited physical resources for development. Efficient cloud resource monitoring can help identify underutilized resources and support appropriate scaling decisions. Releasing unnecessary resources can reduce resource consumption and potentially reduce cloud operating costs and energy usage.

A modular architecture also allows future improvements without completely redesigning the system.

Ethics

The system should use cloud resources responsibly and should not perform unauthorized resource allocation or modification.

Monitoring information should be used only for legitimate infrastructure-management purposes. Access to resource information should be restricted to authorized administrators.

Health & Safety

The proposed system is an infrastructure-management application and does not directly control physical safety-critical equipment. However, poor resource-management decisions could affect the availability and performance of applications running on the cloud infrastructure.

Therefore, monitoring and scaling mechanisms should be tested carefully before deployment.

Security

Cloud infrastructure monitoring can contain sensitive operational information. The system should:

- Restrict monitoring access to authorized users.
- Protect cloud API credentials and access keys.
- Apply suitable authentication and authorization.
- Secure communication between system components.
- Avoid unnecessary storage of sensitive monitoring information.
- Maintain appropriate logs for resource-management activities.

Risk Register

Table 3.5 – Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation
Incorrect monitoring data	Medium	High	High	Validate monitoring metrics
Monitoring service failure	Medium	High	High	Fault detection and recovery mechanisms
Delayed metric collection	Medium	Medium	Medium	Configure suitable monitoring intervals
Incorrect threshold configuration	Medium	High	High	Test and tune threshold values
High monitoring overhead	Low	Medium	Low	Optimize monitoring frequency

Network connectivity failure	Medium	High	High	Retry and recovery mechanisms
Unauthorized monitoring access	Low	High	Medium	Authentication and access control
Exposure of cloud credentials	Low	High	Medium	Secure credential storage
Excessive resource scaling	Medium	High	High	Apply scaling limits and validation
Insufficient scaling	Medium	High	High	Monitor workload and test scaling policies
Underutilized resources not detected	Medium	Medium	Medium	Continuous utilization analysis
Loss of monitoring data	Low	Medium	Low	Logging and backup mechanisms

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The development of Module 3 – Monitoring & Cloud Resource Management follows a modular and systematic approach. The module is designed to continuously monitor the performance and utilization of provisioned cloud resources and provide the required information for resource-management and auto-scaling decisions.

Development Steps

1. Identify the cloud resources and monitoring parameters.
2. Select important metrics such as CPU, memory, network, and storage utilization.
3. Design the monitoring mechanism.
4. Implement resource-metric collection.
5. Implement utilization analysis.
6. Define and configure resource-utilization thresholds.
7. Detect overloaded and underutilized virtual machines.
8. Generate structured monitoring information.
9. Integrate monitoring information with the auto-scaling and resource-management module.
10. Test the monitoring system under different workload conditions.

Resources

Table 4.1 – Development Resources

Resource	Purpose
Python	Monitoring-system development
Cloud Platform	Hosting and managing virtual infrastructure
Cloud Virtual Machines	Resources to be monitored
Python Libraries	Metric collection and data processing
Monitoring Tools/APIs	Collection of cloud resource metrics
CPU/Memory Metrics	Resource-utilization analysis
Network/Storage Metrics	Infrastructure monitoring
Computer/Laptop	Development and testing
Dashboard/Visualization Tools	Displaying monitoring results
Internet Connection	Access to cloud infrastructure

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Python

Python is used as the primary programming language for implementing the monitoring and cloud resource-management functions. It can be used to collect, process, analyse, and display cloud resource utilization information.

Cloud Computing Platform

A cloud platform is used to create and manage virtual infrastructure. The provisioned virtual machines provide the resources that are continuously monitored by Module 3.

Resource Monitoring

The monitoring component collects important infrastructure metrics such as:

- CPU utilization
- Memory utilization
- Network usage
- Storage utilization
- Virtual machine status
- Resource availability

Data Processing

The collected monitoring data is processed to determine the current condition of cloud resources. The

system compares resource utilization with predefined thresholds to identify overloaded or underutilized virtual machines.

Auto Scaling Integration

Module 3 provides monitoring information to the auto-scaling component. When resource utilization reaches a predefined condition, the monitoring system can support decisions such as scaling up or scaling down cloud resources.

Modern Tools Used

The implementation can use modern cloud and development tools such as:

- Python
- Cloud provider APIs
- Virtual machines
- Monitoring APIs
- Cloud dashboards
- Data visualization libraries
- Git/GitHub for source-code management

4.3 Test Plan & Verification

Table 4.2 – Test Plan and Verification

Requirement	Test Method	Acceptance Criterion
Virtual machine monitoring	Start an active VM and monitor it	VM status displayed correctly
CPU monitoring	Generate different CPU workloads	CPU utilization collected correctly
Memory monitoring	Increase memory usage	Memory utilization displayed
Network monitoring	Generate network traffic	Network usage metrics collected
Storage monitoring	Add/remove files from VM storage	Storage utilization updated
High utilization detection	Increase CPU/memory above threshold	High-utilization condition identified
Low utilization detection	Keep VM workload below threshold	Underutilized condition identified
Threshold evaluation	Test values below and above threshold	Correct condition reported
Metric collection	Collect metrics at defined intervals	Monitoring data updated regularly
Dashboard	View collected monitoring information	Data displayed clearly
Auto-scaling integration	Trigger scaling condition	Scaling information generated
Multiple VMs	Monitor multiple virtual machines	Resources monitored correctly
Failure handling	Temporarily interrupt monitoring connection	Failure identified and handled

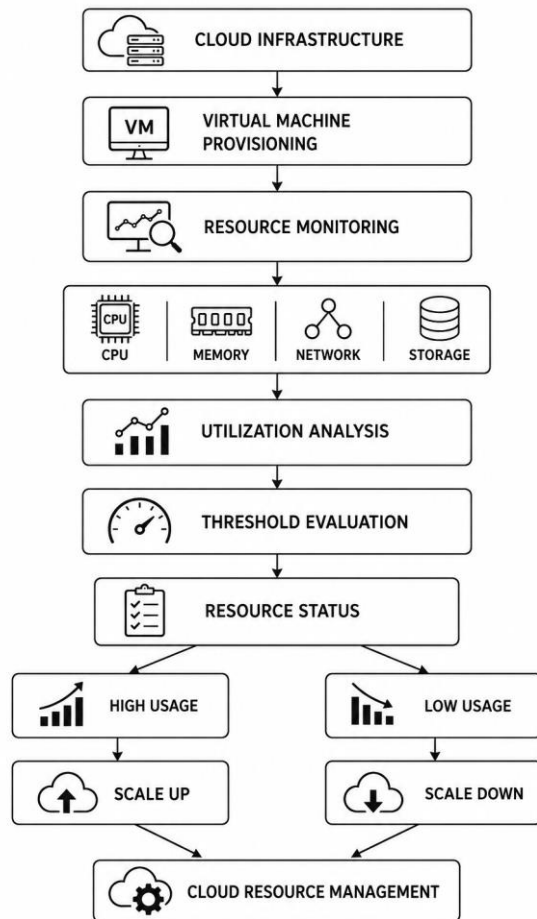
4.4 Results

The expected output of Module 3 is:

Continuous Cloud Resource Monitoring and Resource-Management Information

The module successfully monitors provisioned virtual infrastructure and provides information about current resource utilization.

Fig No 1.4 System Flow Diagram for Cloud Resource Management and Auto-Scaling



The expected monitoring output may include:

Resource	Utilization	Status
CPU	Monitored %	Normal / High
Memory	Monitored %	Normal / High
Network	Monitored usage	Normal / High
Storage	Monitored %	Normal / High
Virtual Machine	Active/Inactive	Available/Unavailable

4.5 Data Analysis & Engineering Judgment

The quality and timeliness of monitoring data directly influence the effectiveness of cloud resource-management decisions.

A well-designed monitoring stage provides:

- Continuous resource visibility.
- Accurate utilization information.
- Early identification of overloaded resources.
- Identification of underutilized resources.
- Better auto-scaling decisions.
- Improved resource utilization.
- Reduced unnecessary resource consumption.
- Better cloud infrastructure performance.

The engineering decision to use continuous automated monitoring allows the system to observe changing workloads without requiring constant manual intervention.

Threshold-based monitoring is selected because it provides a simple and understandable mechanism for identifying resource conditions. For example, when CPU utilization exceeds a predefined threshold, the system can indicate a possible need for additional resources. When utilization remains

low, the system can indicate an opportunity for resource reduction.

The monitoring module is designed independently so that it can be tested before complete integration with the auto-scaling mechanism.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 4.3 – Achievement of Objectives

Objective	Evidence	Achievement
Identify cloud resources to be monitored	Virtual machines and cloud resources	Fully Achieved
Monitor CPU utilization	CPU monitoring component	Fully Achieved
Monitor memory utilization	Memory monitoring component	Fully Achieved
Monitor network usage	Network metrics	Achieved
Monitor storage utilization	Storage metrics	Achieved
Detect overloaded resources	Threshold evaluation	Fully Achieved
Detect underutilized resources	Low-utilization detection	Fully Achieved
Provide structured monitoring data	Monitoring records/metrics	Fully Achieved
Support auto-scaling decisions	Scaling-condition information	Fully Achieved
Provide scalable architecture	Modular monitoring design	Achieved

4.7 Strengths & Limitations

Strengths

- Continuous monitoring of cloud resources.
- Automated collection of resource metrics.
- Easy identification of high resource utilization.
- Detection of underutilized resources.
- Supports auto-scaling decisions.
- Improves visibility into virtual infrastructure.
- Modular and scalable architecture.
- Reduces dependency on manual monitoring.
- Can support multiple virtual machines.
- Can be extended with additional monitoring metrics.

Limitations

- Depends on reliable cloud monitoring services and APIs.
- Monitoring itself may introduce a small amount of resource overhead.
- Incorrect threshold configuration can result in inappropriate scaling decisions.
- Network failures may temporarily interrupt metric collection.
- Large cloud environments can generate significant monitoring data.
- Advanced predictive monitoring may require historical workload data.
- Cloud-provider-specific monitoring features may limit portability.

4.8 Project Planning, Roles & Budget

Project Timeline

Phase	Activity
Phase 1	Problem identification
Phase 2	Literature review
Phase 3	Identification of cloud resources and metrics
Phase 4	Monitoring module design
Phase 5	Cloud environment setup
Phase 6	Python monitoring implementation
Phase 7	Threshold and utilization analysis
Phase 8	Integration with auto-scaling
Phase 9	Testing and verification
Phase 10	Results analysis

Phase 11	Documentation
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Team Roles

Role	Responsibility
Team Member 1	Cloud infrastructure setup
Team Member 2	Monitoring implementation
Team Member 3	Resource-utilization analysis
Team Member 4	Testing and documentation

Budget

Since the module is primarily software-based, the main resources are a computer, internet connection, development software, and cloud infrastructure.

Item	Estimated Cost
Computer/Laptop	Existing resource
Python	Free/Open-source
Development Tools	Free/Open-source
Cloud Platform	Free tier / Educational resources
Monitoring Tools	Free/Open-source or cloud-provided
Internet	Existing resource
Total Additional Cost	Minimal

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

Module 3 – Monitoring & Cloud Resource Management provides an important foundation for the Cloud-Based IaaS Virtual Infrastructure Provisioning and Auto Scaling System. The module continuously monitors important cloud resource parameters including CPU utilization, memory utilization, network usage, storage usage and virtual machine status. It collects and analyzes resource information to identify overloaded and underutilized virtual machines.

The quality and timeliness of monitoring data are important because they directly affect resource-management and auto-scaling decisions. By using a structured and modular approach, the proposed system provides a clear connection between cloud resource monitoring, utilization analysis, threshold evaluation and resource allocation.

The complete project combines virtual infrastructure provisioning, resource monitoring, utilization analysis and auto-scaling mechanisms. The system is intended to improve cloud resource utilization, reduce unnecessary resource consumption and maintain better application performance by dynamically managing virtual infrastructure according to workload conditions.

5.2 Future Work

Module 3 can be improved in future versions by:

1. Adding more cloud resource monitoring parameters.
2. Providing real-time monitoring with shorter monitoring intervals.
3. Developing advanced cloud monitoring dashboards.
4. Integrating predictive analytics for workload forecasting.
5. Using machine learning for predictive auto-scaling.

6. Supporting multiple cloud platforms and hybrid-cloud environments.
7. Adding advanced cost and energy-efficiency monitoring.
8. Improving fault detection and automatic recovery mechanisms.
9. Supporting larger numbers of virtual machines and cloud resources.
10. Integrating centralized monitoring and alert notification systems.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned how to design a modular cloud resource monitoring system.
- Learned the importance of CPU, memory, network and storage monitoring.
- Understood the concepts of IaaS and virtual infrastructure management.
- Learned how resource utilization can be analyzed using monitoring data.
- Understood the relationship between monitoring and auto-scaling.
- Learned how threshold-based decisions can support cloud resource management.
- Understood the importance of continuous monitoring in cloud environments.

Engineering & Professional Skills Developed

- Problem identification.
- Cloud system design.
- Python programming.
- Virtual machine management.
- Resource monitoring.
- Data organization and analysis.
- Threshold configuration.
- Testing and debugging.
- Technical documentation.
- Team collaboration.

Individual Reflection

During the development of Module 3, the main engineering challenge was designing an effective monitoring mechanism that could continuously collect and analyze cloud resource information. The importance of accurate and timely monitoring became clear because incorrect or delayed resource information can result in inefficient resource allocation and inappropriate scaling decisions.

The continuous automated monitoring approach was selected because it provides better visibility of virtual machine performance compared with manual monitoring. Monitoring multiple parameters such as CPU utilization, memory utilization, network usage and storage usage also provides a more complete view of the cloud infrastructure.

Through this module, knowledge of cloud computing, IaaS, virtual machines, Python programming, resource monitoring and auto-scaling workflows was improved. If the project were redesigned, predictive monitoring, machine-learning-based workload forecasting, advanced dashboards and multi-cloud resource management would be considered to improve the scalability and efficiency of the system.

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APPENDICES

APPENDIX A – DETAILED CALCULATIONS

For Module 3 – Monitoring & Cloud Resource Management, the system mainly collects and analyzes cloud resource utilization data. Basic calculations can be used to determine resource utilization and identify overloaded or underutilized virtual machines.

A.1 CPU Utilization Calculation

CPU utilization can be calculated as:

$$\text{CPU Utilization (\%)} = (\text{Used CPU} / \text{Allocated CPU}) \times 100$$

Example:

If:

- Allocated CPU = 4 vCPUs
- Used CPU = 3 vCPUs

Then:

$$\text{CPU Utilization} = (3 / 4) \times 100$$

$$\text{CPU Utilization} = 75\%$$

The calculated value can be compared with the configured threshold to determine whether additional resources are required.

A.2 Memory Utilization Calculation

Memory utilization can be calculated as:

$$\text{Memory Utilization (\%)} = (\text{Used Memory} / \text{Allocated Memory}) \times 100$$

Example:

If:

- Allocated Memory = 8 GB
- Used Memory = 6 GB

Then:

$$\text{Memory Utilization} = (6 / 8) \times 100$$

$$\text{Memory Utilization} = 75\%$$

A.3 Resource Utilization Status

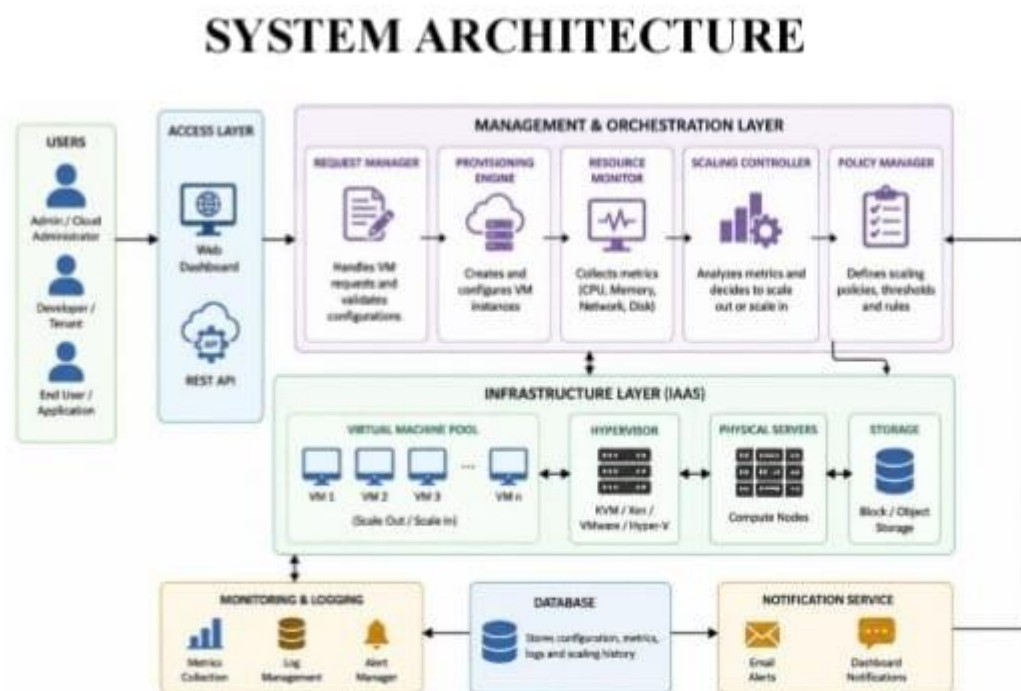
A simple threshold-based classification can be used:

- High Utilization: Utilization > 80%
- Normal Utilization: Utilization between 40% and 80%
- Low Utilization: Utilization < 40%

These values are example thresholds and can be modified according to the requirements of the cloud environment.

APPENDIX B – SYSTEM WORKFLOW

Monitoring & Cloud Resource Management Workflow



Module 3 Process

START

↓

Connect to Cloud Infrastructure

↓

Identify Virtual Machines

↓

Collect Resource Metrics

↓

Monitor CPU, Memory, Network and Storage

↓

Analyze Resource Utilization

↓

Compare Utilization with Threshold

↓

Is Utilization High?

↓

YES → Scale Up / Allocate Additional Resources

NO → Check for Low Utilization

↓

Low Utilization → Scale Down / Release Unused Resources

↓

Update Resource Status

↓

Display Monitoring Information

↓

END

APPENDIX C – SAMPLE RESOURCE MONITORING CODE

```
print("CLOUD-BASED IaaS SYSTEM")
```

```
print("MODULE 3 - MONITORING & CLOUD RESOURCE MANAGEMENT")
```

```
cpu = float(input("Enter CPU Utilization (%): "))
```

```
memory = float(input("Enter Memory Utilization (%): "))
```

```
network = float(input("Enter Network Usage (%): "))
```

```

storage = float(input("Enter Storage Utilization (%): "))

if cpu < 0 or cpu > 100:

    print("Invalid CPU utilization")

elif memory < 0 or memory > 100:

    print("Invalid memory utilization")

elif network < 0 or network > 100:

    print("Invalid network usage")

elif storage < 0 or storage > 100:

    print("Invalid storage utilization")

else:

    print("\nResource monitoring completed")


print("CPU:", cpu, "%")

print("Memory:", memory, "%")

print("Network:", network, "%")

print("Storage:", storage, "%")

if cpu > 80 or memory > 80:

    print("\nStatus: HIGH UTILIZATION")

    print("Action: Scale Up")

elif cpu < 40 and memory < 40:

    print("\nStatus: LOW UTILIZATION")

    print("Action: Scale Down")

else:

    print("\nStatus: NORMAL UTILIZATION")

    print("Action: Maintain Current Resources")

```

APPENDIX D – CLOUD RESOURCE MONITORING SPECIFICATION

Resource Monitoring Parameters

S.No.	Parameter	Data Type	Purpose
1	CPU Utilization	Float	Measures CPU usage of the VM
2	Memory Utilization	Float	Measures memory usage
3	Network Usage	Float	Measures network resource usage
4	Storage Utilization	Float	Measures storage usage
5	VM Status	String	Indicates VM operational status
6	Monitoring Time	Date/Time	Records the time of measurement

Resource Validation Specification

Parameter	Validation
CPU Utilization	Numerical value between 0–100%
Memory Utilization	Numerical value between 0–100%
Network Usage	Numerical value between 0–100%
Storage Utilization	Numerical value between 0–100%
VM Status	Valid operational status
Monitoring Time	Valid date and time

Threshold Specification

Resource	Low	Normal	High
CPU	< 40%	40–80%	> 80%
Memory	< 40%	40–80%	> 80%
Network	< 40%	40–80%	> 80%
Storage	< 40%	40–80%	> 80%

APPENDIX E – SAMPLE MONITORING TEST DATA

Test ID	CPU (%)	Memory (%)	Network (%)	Storage (%)	Resource Status	Recommended Action
T01	45	50	40	55	Normal	Maintain
T02	85	82	65	70	High	Scale Up
T03	30	35	28	40	Low	Scale Down
T04	75	70	60	65	Normal	Maintain
T05	92	88	80	75	High	Scale Up
T06	25	30	35	32	Low	Scale Down

Purpose of Monitoring Data

The data is used to verify whether:

- Cloud resources are monitored correctly.
- CPU utilization is collected correctly.
- Memory utilization is identified correctly.

- Network and storage usage are monitored.
- High resource utilization is detected.
- Low resource utilization is detected.
- Threshold conditions are evaluated correctly.
- Appropriate scaling actions can be identified.
- Monitoring information is prepared for resource management.

Important: These are sample test values for demonstrating the appendix, not claimed experimental results. Replace them with your actual monitoring results before submission.

APPENDIX F – ETHICAL, SECURITY AND SAFETY CONSIDERATIONS

The cloud-based system manages virtual infrastructure and resource information. Therefore, responsible cloud resource management and secure access are important.

The following principles should be followed:

1. Cloud resources should be accessed only by authorized users.
2. Cloud credentials and API keys should be protected.
3. Monitoring information should not be unnecessarily exposed.
4. Resource information should be used only for the intended project purpose.
5. Secure communication should be used between the monitoring application and cloud services.
6. Access permissions should follow the principle of least privilege.
7. Monitoring data should be protected against unauthorized modification.
8. Auto-scaling decisions should be monitored to prevent unnecessary resource allocation.
9. Resource management should consider cost and energy efficiency.
10. System failures should be handled without unnecessarily affecting application availability.

Institutional Approval

If institutional approval is required, include:

- Approval document
- Approval number
- Date of approval
- Faculty/department authorization

If institutional approval is not applicable:

Ethical/institutional approval was not applicable to this software-based academic project because the project does not involve human subjects, clinical intervention, or direct physical experimentation.

APPENDIX G – PROJECT PROGRESS RECORD

Meeting	Date	Discussion	Work Completed	Next Activity
1	—	Project topic selection	Topic finalized	Literature review
2	—	Project requirements	Requirements identified	Module design

3	_____	Module 3 discussion	Monitoring parameters identified	Monitoring implementation
4	_____	Cloud environment setup	Virtual infrastructure configured	Resource monitoring
5	_____	Python implementation	Initial monitoring code developed	Threshold configuration
6	_____	Testing	Test cases prepared	Error correction
7	_____	Module integration	Monitoring connected with resource management	Final testing
8	_____	Results analysis	Monitoring results documented	Report preparation
9	_____	Report preparation	Documentation completed	Final submission

Meeting Discussion Points

During project meetings, the following points can be recorded:

- Selection of cloud-based IaaS as the project topic.
- Identification of virtual infrastructure requirements.
- Selection of resource monitoring parameters.
- Design of Module 3.
- Selection of Python for implementation.
- Cloud environment configuration.
- CPU, memory, network and storage monitoring.
- Threshold selection.
- Resource management and auto-scaling integration.
- Testing and debugging.
- Results analysis and documentation.

APPENDIX I – FEEDBACK FORM

S.No.	Evaluation Area	Rating / Feedback
1	Ease of understanding the monitoring dashboard	_____
2	Clarity of resource utilization information	_____
3	Accuracy of monitoring information	_____
4	Understanding of scaling conditions	_____
5	Overall usability	_____
6	Suggestions for improvement	_____

Sample Feedback Questions

1. Is the resource monitoring information easy to understand?
2. Are CPU and memory utilization clearly displayed?
3. Are network and storage parameters clearly presented?
4. Is the high/low utilization status understandable?

5. Are the scaling actions easy to understand?
6. What additional monitoring parameters should be included?
7. What improvements would make the monitoring system more useful?

APPENDIX J – INDIVIDUAL CONTRIBUTION EVIDENCE

Student	Responsibility	Work Done	Contribution %
Student 1	Module 1	Virtual infrastructure provisioning	___%
Student 2	Module 2	Auto-scaling and resource allocation	___%
Student 3	Module 3	Monitoring and cloud resource management	___%
Student 4	Testing & Documentation	Testing and report preparation	___%
Total			100%

Module 3 Individual Contribution

If you are specifically responsible for Module 3 – Monitoring & Cloud Resource Management, your contribution can be described as:

- Identified the cloud resources that require monitoring.
- Selected CPU, memory, network and storage utilization parameters.
- Designed the monitoring workflow.
- Implemented resource monitoring using Python and cloud APIs/tools.
- Implemented resource utilization analysis.
- Configured threshold conditions for high and low utilization.
- Identified overloaded and underutilized virtual machines.
- Supported resource-management and auto-scaling decisions.
- Tested monitoring using different workload conditions.
- Analyzed monitoring results.
- Prepared documentation related to Module 3.

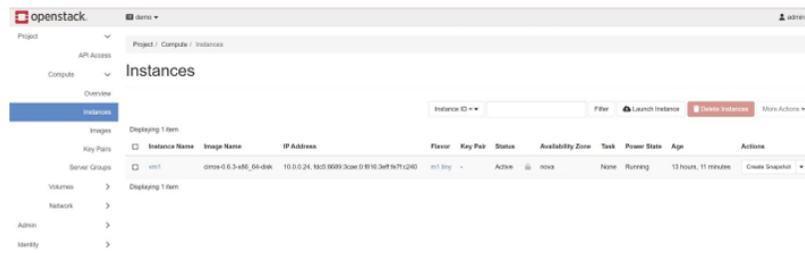


Figure 3.1: Horizon Monitoring Dashboard.

```
divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack$ openstack compute service list
```

ID	Binary	Host	Zone	Status	State	Updated At
6130d750-78d2-41b1-b064-22a2ebbc68eb	nova-scheduler	LAPTOP-EDH45F4T	internal	enabled	up	2026-08-20T05:28:47.000000
32a23a33-9871-4a00-ae94-2bc8f9caad0b	nova-conductor	LAPTOP-EDH45F4T	internal	enabled	up	2026-08-20T05:28:47.000000
74377916-8a31-4d0b-9da2-6529c5ae6cbf	nova-conductor	LAPTOP-EDH45F4T	internal	enabled	up	2026-08-20T05:29:11.000000
cb3e7d3b-0b1b-4ee4-98a6-30d51a6321de	nova-compute	LAPTOP-EDH45F4T	nova	enabled	up	2026-08-20T05:28:48.000000

Figure 3.2: OpenStack Compute Services Status.

```
divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack$ openstack server list --all-projects
```

ID	Name	Status	Networks	Image	Flavor	Project ID
c448dca4-7c8a-4878-8ee8-dfb11ded7838	vm1	ACTIVE	private=10.0.0.24, fdc5:8689:3cae:0:f816:3eff:fe7f:c240	N/A (booted from volume)	m1.tiny	3a38c3c2ff4b4d53a80a37653a047f6a

Figure 3.3: OpenStack Service Monitoring.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)